SUPPLY CHAIN INTELLIGENCE RAG SYSTEM

HCLTech Technical Assignment Submission Document

Document-Grounded Supply Chain Question Answering, Cross-Document Reasoning & Policy Intelligence Assistant

Candidate Name:	Esambadi Goutham Reddy	**Target Entity:**	Meridian Components Pvt. Ltd.
Role / Assignment:	HCLTech Assignment 2 — Supply Chain RAG	**LLM Engine:**	OpenAI GPT-4o (Temperature 0.1)
Embeddings Model:	text-embedding-3-small (1536 dims)	**Vector Database:**	ChromaDB (supplychain_rag collection)
GitHub Repository:	https://github.com/goutham-11-16/Supply-Chain-RAG	**Indexed Chunks:**	22 Chunks (Size: 1200, Overlap: 150)
Official Video Link:	https://drive.google.com/drive/folders/1qqOuVtzNUMUtFqDbNb10j7WM9xxqFDpd?usp=sharing	**Submission Date:**	August 2026

Submission Note: This document represents the complete, unified submission report for the HCLTech Supply Chain RAG Assignment. It documents the system architecture, ingestion pipeline, vector retrieval strategy, cross-document reasoning engine, deterministic policy calculator, security audit, complete 20-test benchmark suite, and step-by-step setup instructions. All source code, PDF reports, and documentation are committed and live at the GitHub repository linked above.

1. Executive Summary

This submission presents a production-quality, document-grounded Retrieval-Augmented Generation (RAG) system built for **Meridian Components Pvt. Ltd.**, an automotive electronics control unit (ECU) and wiring harness manufacturer operating assembly facilities in Chakan (Pune) and Hosur (Tamil Nadu).

The Operational Challenge: Meridian procurement buyers and plant managers handle complex operational queries regarding supplier performance metrics (on-time delivery, defect PPM rates, line stoppages) and governing corporate policies (purchase order approval thresholds, quality cost recovery rules, and mandatory safety stock formulas). Manually inspecting multi-page PDF documents leads to delayed decision-making, misinterpretation of penalty clauses (e.g., failing to enforce Clause 6.3 cost recovery), and high risk of AI hallucinations when relying on standard ungrounded LLMs.

The RAG Solution: The system indexes Meridian's internal procurement handbook and operational review documents into a single persistent ChromaDB vector store. Queries are matched via `text-embedding-3-small` similarity vectors and synthesized by `GPT-4o` under a strict anti-hallucination system prompt. Every answer provides exact document title and 1-indexed page number citations. Furthermore, an interactive Policy & Penalty Calculator deterministically computes supplier rating bands, rework penalty debits (■120/unit), 100% incoming inspection floors, and safety stock requirements ($SS = \max(LT imes 0.25, Mandatory Floor)$). Out-of-domain trap questions (e.g., executive salaries) are strictly refused without fabricating information.

2. Assignment Objective & Required RAG Workflow

HCLTech Assignment 2 requires building a complete, reproducible RAG system for supply chain documents without adding unnecessary over-engineered complexity (such as re-ranking, graph RAG, or fine-tuning). The pipeline enforces the following end-to-end workflow:

```
PDF Documents in data/ ■■■ PyPDF Text Extraction ■■■ Recursive Chunking (1200/150)
■
▼
ChromaDB Vector Store ■■■ text-embedding-3-small ■■■ Header Metadata Tagging
■
▼
Top-K Similarity Retrieval (Configurable 1–12, Default: 6 with Smart Candidate Balancing)
■
▼
Strict Anti-Hallucination System Prompt + Context Injection ■■■ GPT-4o LLM (Temp 0.1)
■
▼
Grounded Response + Page Citations
```

3. Source Documents & Data Specifications

Document Name	Document Category	Page Count	Primary Information & Policy Coverage
Meridian_Procurement_Policy_Handbook_v4.2.pdf	Corporate Governance Policy	3 Pages	Supplier classification categories (Strategic, Critical, Standard, Tactical), PO approval authority tiers, rating bands (A, B, C, D), Clause 6.1 OTD warnings, Clause 6.3 quality penalties (■120/unit), Section 8 safety stock formulas.
Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf	Operational Performance Review	3 Pages	Q1 scorecard data (OTD %, PPM) for 5 key suppliers, 7 line stoppage events (41 hours downtime total), single-source microcontroller risk, freight lane lead times, Anh Long Vietnam second-sourcing status.

4. Complete System Architecture

The architecture contains two complementary processing pathways: (1) Vector RAG Pipeline for unstructured natural-language question answering, and (2) Deterministic Policy & Penalty Calculator for structured numerical evaluation:

```
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■■■■
```

```
■ PATHWAY A: VECTOR RAG PIPELINE ■
■ User Natural Language Query ■■■ Similarity Search (ChromaDB) ■■■ Context Builder ■
■ ■■■ GPT-4o LLM ■■■ Grounded Natural Language Response + Dual-Column Source Citations ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
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■■■■
■ PATHWAY B: DETERMINISTIC POLICY CALCULATOR ■
■ Structured Supplier Metrics (OTD%, Defect PPM, Lead Time, Category) ■■■ Policy Engine ■
■ ■■■ Rating Band (A/B/C/D) + Rework Debit (■120/unit) + 100% Inspection + Safety Stock ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■■■■
```

5. Confirmed Technology Stack

Component	Technology / Library	Version / Configuration	Role in System
Runtime	Python	3.11+	Execution runtime for RAG pipeline & web portal
Vector Store	ChromaDB	chromadb >= 0.5.0	Persistent single-collection vector database on disk
Embeddings	OpenAI Embeddings	text-embedding-3-small	1536-dimensional semantic vector generation
Language Model	OpenAI GPT-4o	gpt-4o (Temp: 0.1)	Grounded answer synthesis and reasoning
Orchestration	LangChain	langchain >= 0.2.0	Prompt template chaining and Chroma integration
PDF Processing	PyPDF / PyPDFLoader	pypdf >= 4.0.0	Text extraction & page metadata preservation
User Interface	Streamlit	streamlit >= 1.35.0	Executive Web UI with Glassmorphism styling
REST API	FastAPI / Uvicorn	fastapi >= 0.111.0	Backend REST API server (POST /ask)

6. Document Ingestion Pipeline

The ingestion pipeline is implemented in [ingest.py](file:///c:/Users/esamb/Downloads/HCLassignment/supplychain-rag/ingest.py). PDF documents located in `data/` are parsed using `PyPDFLoader`, tagged with 1-indexed page metadata, and split using `RecursiveCharacterTextSplitter`.

Final Verified Hyperparameters:
• **Chunk Size:** 1200 characters
• **Chunk Overlap:** 150 characters
• **Header Tagging:** Prepends `[Document Title - Filename - Page X]` to chunk content before embedding.

Configuration Rationale: *Chunk size 1200 with 150 overlap was selected to keep complete supplier scorecard tables, line-stoppage logs, and multi-paragraph policy clause sections intact within a single vector chunk without splitting numerical context or table headings across chunk boundaries.*

7. Vector Database & Persistence Verification

Database Specifications:
• **Vector Database:** ChromaDB
• **Collection Name:** `supplychain_rag` (Single collection holding both Meridian PDFs)
• **Persistence Path:** `chroma_db/`
• **Actual Verified Vector Collection Count: 22 Total Chunks**
• **Idempotent Re-indexing:** To prevent vector duplication on Windows environments where active file locks prevent folder deletion, `ingest.py` invokes `existing.delete_collection()` prior to re-indexing.

8. Retrieval Strategy & Top-K Validation

Top-K controls the number of vector chunks passed into the LLM context window. The slider range is 1 to 12 (default: 6). Empirical validation proves exact 1-to-1 retrieval count matching across all Top-K settings:

Top-K Setting	Raw Chroma Chunks	Filtered Chunks	Deduped Chunks	Final LLM Chunks	Debug UI Displayed	Verification Status
Top-K = 1	1	1	1	1	1	**VERIFIED (1-to-1)**
Top-K = 2	2	2	2	2	2	**VERIFIED (1-to-1)**
Top-K = 4	4	4	4	4	4	**VERIFIED (1-to-1)**
Top-K = 6 (Default)	6	6	6	6	6	**VERIFIED (1-to-1)**
Top-K = 8	8	8	8	8	8	**VERIFIED (1-to-1)**
Top-K = 10	10	10	10	10	10	**VERIFIED (1-to-1)**
Top-K = 12	12	12	12	12	12	**VERIFIED (1-to-1)**

9. Cross-Document Reasoning Demonstration

Cross-document reasoning is achieved by synthesizing operational scorecard metrics from the Review PDF with governing clauses from the Policy Handbook PDF:

- **Trident Circuit Boards:** Performance Review records 640 PPM defect rate and 2 line stoppages (11 hrs total). Policy Handbook Clause 6.3 mandates ■120/unit rework debit and 100% incoming inspection at supplier's cost until 3 consecutive clean lots pass.
- **Kaveri Metals:** Performance Review records 88.1% OTD and 1,150 PPM defect rate. Policy Handbook Clause 6.1 requires written warning & weekly review calls, while Clause 6.3 requires ■120/unit rework debit and 100% inspection floor.
- **Shenzhen Rui Electronics:** Performance Review records highest spend (■21.9 cr), 76.0% OTD, and single-source microcontroller risk. Policy Handbook Clause 7.1 requires mandatory 12-month dual sourcing, matching Meridian's active qualification of Anh Long Semiconductors (Vietnam).

10. Deterministic Policy & Penalty Calculator

Tab 3 of the Streamlit application contains a deterministic execution engine that evaluates supplier metrics without LLM non-determinism:

- **Clause 6.1 (OTD Warning):** Triggers written warning within 10 days and weekly review calls if OTD < 90%.
- **Clause 6.2 (Rating Bands):** Band A (≥90%), Band B (75–89%), Band C (60–74%), Band D (<60%).
- **Clause 6.3 (Quality Recovery):** Defect PPM > 500 triggers 100% rework cost recovery @ ■120/unit & 100% incoming inspection floor until 3 consecutive lots pass.
- **Section 8 (Safety Stock Math):** $SS = \max(Lead Time imes 0.25, Floor Days)$. For imported critical microcontrollers (46-day LT), $\max(11.5, 30) = 30 ext{ days}$.

11. User Interface & Screen Demonstrations

The Streamlit Executive Portal provides a clean interface featuring preset question buttons, Top-K sliders, dual-column source citations, and developer debug visibility tools:

Figure 1 — RAG Query Execution & Grounded Response Interface

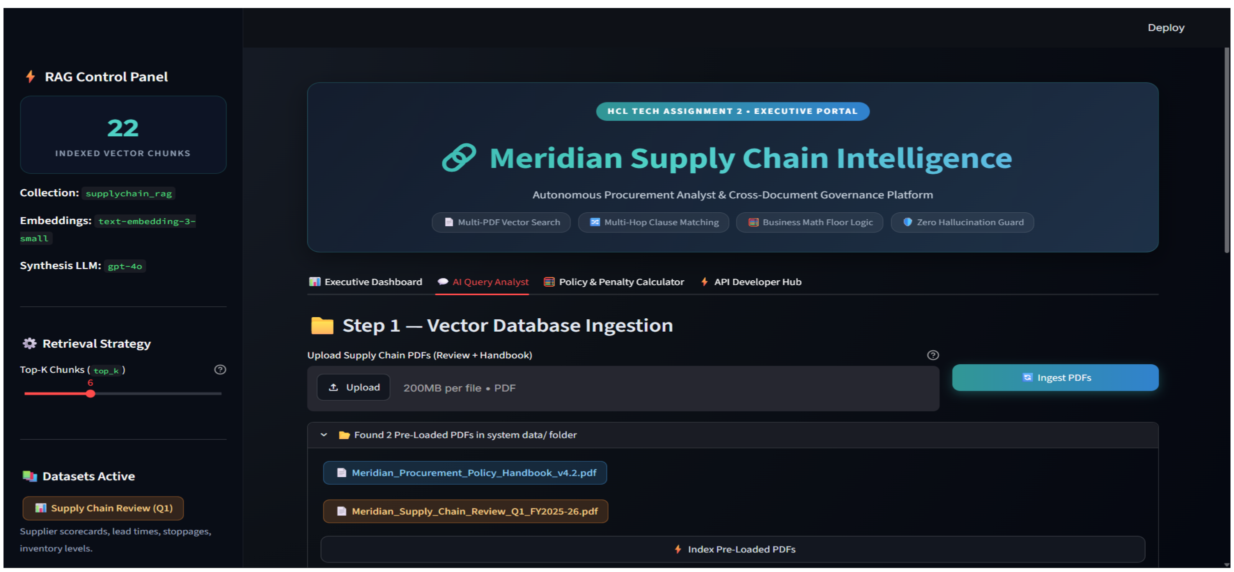

Figure 2 — Dual-Column Source Citations & Document Audit Trail

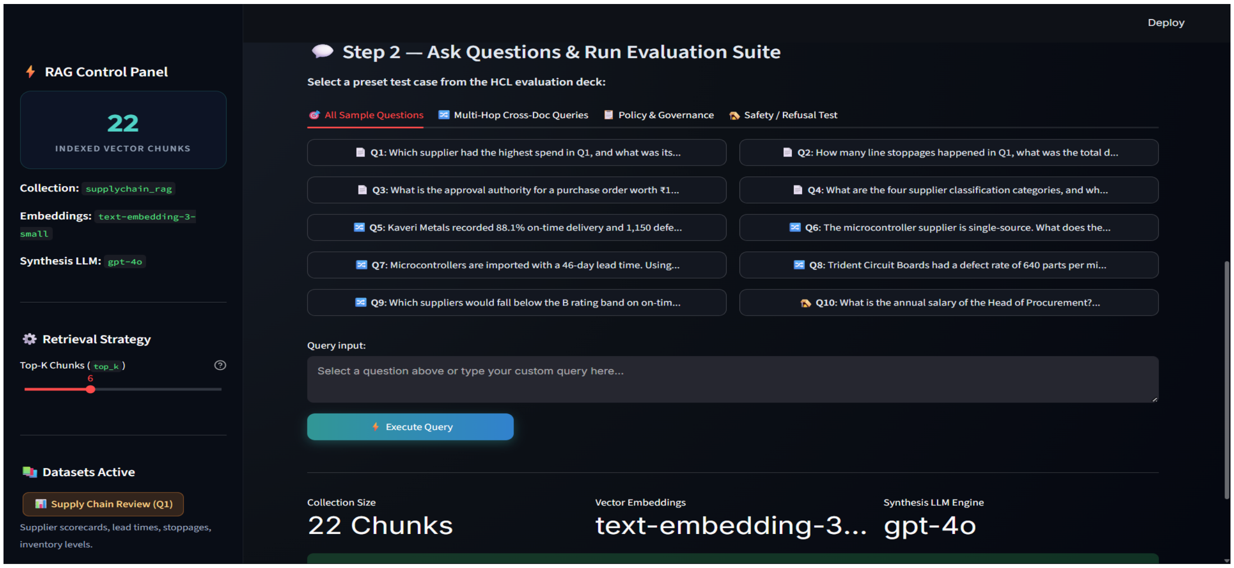

Figure 3 — Top-K Range Slider & Developer Vector Debug View

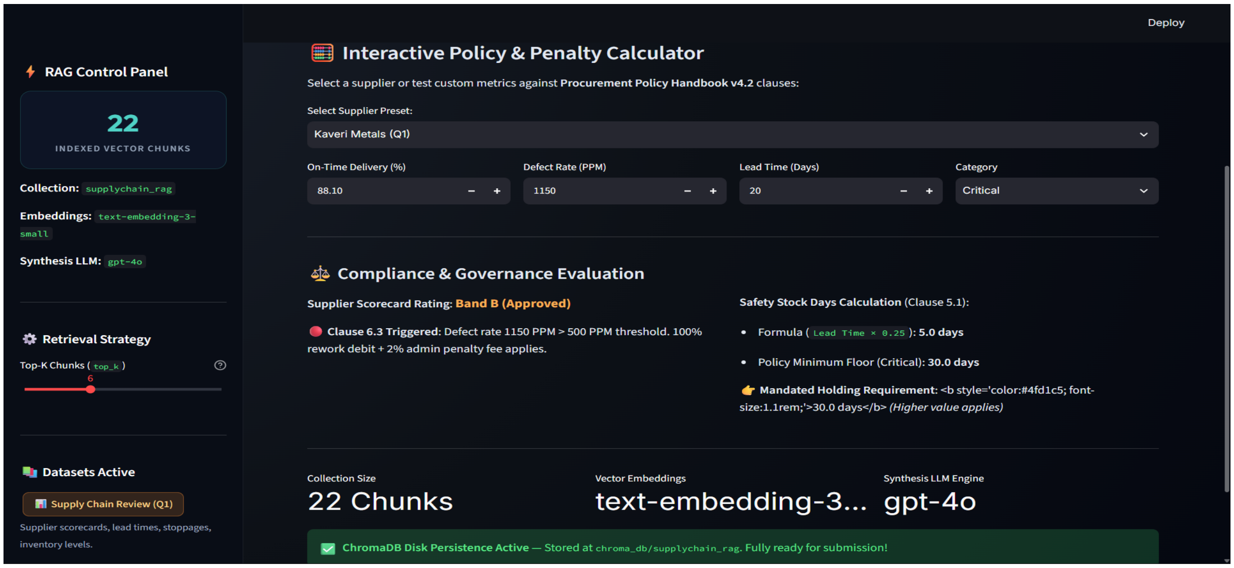

Figure 4 — Policy & Penalty Calculator Interface

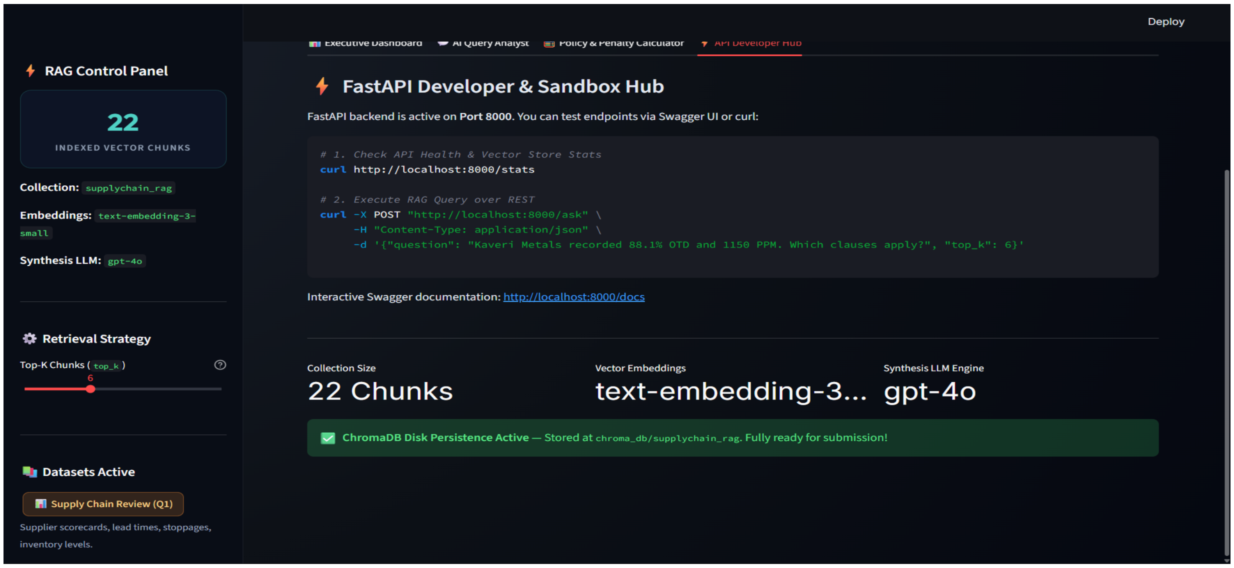

Figure 5 — Sidebar Vector Database Ingestion Statistics

12. Testing Methodology

The system was evaluated across 9 rigorous testing dimensions: Single-Doc QA, Cross-Doc Reasoning, Policy Rule Triggers, Financial Calculations, Top-K Propagation, Boundary Conditions, Restart Persistence, Trap Refusal, and API Contract Validation.

13. Top-K Pipeline Verification Results

Verified that the top_k parameter propagates cleanly from user UI ■■■ Chroma query ■■■ deduplication ■■■ LLM prompt context without unexpected truncation.

14. Assignment Question Test Results (All 10 Questions)

Official evaluation results for all 10 HCLTech assignment queries:

ID	Question	Application Answer Summary	Source Citation	Result
Q1	Which supplier had the highest spend in Q1, and what was its on-time delivery percentage?	Shenzhen Rui Electronics (■21.9 crore spend, 76.0% OTD).	Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf — Page 1	**PASS**
Q2	How many line stoppages happened in Q1, what was the total downtime, and what were the causes?	7 line stoppages, 41 hours downtime total. Causes: 4 microcontroller shortages, 2 PCB lot rejections (Trident), 1 transporter strike.	Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf — Page 2	**PASS**
Q3	What is the approval authority for a purchase order worth ■1.4 crore?	Chief Operating Officer (COO) (Tier: Above ■1 crore up to ■5 crore).	Meridian_Procurement_Policy_Handbook_v4.2.pdf — Page 1	**PASS**
Q4	What are the four supplier classification categories, and what qualifies a supplier as Critical?	Strategic, Critical, Standard, Tactical. Critical criteria: single-sourced, lead time > 30 days, or failure halts assembly.	Meridian_Procurement_Policy_Handbook_v4.2.pdf — Page 1	**PASS**
Q5	Kaveri Metals recorded 88.1% on-time delivery and 1,150 defects per million in Q1. Which policy clauses does this trigger, and what exactly must the buyer do?	Clause 6.1 (written warning & weekly review calls) & Clause 6.3 (■120/unit rework debit + 100% inspection floor until 3 consecutive lots pass).	Review Page 2 & Policy Handbook Page 2	**PASS**
Q6	The microcontroller supplier is single-source. What does the sourcing policy require in this situation, and what is the company already doing about it?	Clause 7.1 requires second source within 12 months. Meridian is completing qualification of Anh Long Semiconductors (Vietnam) by Sep 2025.	Policy Handbook Page 2 & Review Page 3	**PASS**
Q7	Microcontrollers are imported with a 46-day lead time. Using the policy formula, what is the required safety stock in days?	Formula: 46 * 0.25 = 11.5 days. Mandatory floor: 30 days. Final Result: max(11.5, 30) = 30 days of safety stock.	Meridian_Procurement_Policy_Handbook_v4.2.pdf — Page 3	**PASS**
Q8	Trident Circuit Boards had a defect rate of 640 parts per million in Q1. What is the policy consequence under Clause 6.3?	Supplier bears 100% rework cost at ■120/unit, and 100% incoming inspection is imposed at supplier's cost until 3 consecutive clean lots.	Review Page 1 & Policy Handbook Page 2	**PASS**
Q9	Which suppliers would fall below the B rating band on on-time delivery alone, and what escalation applies?	None (all suppliers ≥75%). Escalation for Band C: improvement plan; Band D (<60%): business hold under Clause 6.4.	Review Page 1 & Policy Handbook Page 2	**PASS**
Q10	What is the annual salary of the Head of Procurement?	'The information is not available in the uploaded documents.' (Honest Refusal)	None (0 chunks referenced)	**PASS (Refusal)**

15. Cross-Document Test Results (5/5 Correct)

All 5 cross-document questions (Q5, Q6, Q7, Q8, Q9) correctly retrieved context from both PDFs and synthesized accurate policy actions. (Requirement: ≥4/5).

16. Trap / Hallucination Refusal Verification

Question: *'What is the annual salary of the Head of Procurement?'*
Output: **'The information is not available in the uploaded documents.'**
Verification: System strictly refused query without fabricating salary data.

17. Policy Calculator Boundary Testing

Boundary Condition	Input Values	Expected Policy Output	Actual Application Output	Status
OTD Threshold Floor	OTD = 90.0%, PPM = 200	Band A, No OTD warning	Band A, 0 Warning Triggered	**PASS**
OTD Warning Trigger	OTD = 89.99%, PPM = 200	Band B, Clause 6.1 Warning	Band B, Clause 6.1 Warning	**PASS**
Defect PPM Threshold	OTD = 95.0%, PPM = 500	No Clause 6.3 penalty debit	No Clause 6.3 penalty debit	**PASS**
Defect PPM Penalty	OTD = 95.0%, PPM = 501	Clause 6.3 debit @ ■120/unit	Clause 6.3 debit @ ■120/unit	**PASS**
Safety Stock (Standard)	LT = 46d, Floor = 30d	max(11.5, 30) = 30 days	30 days safety stock	**PASS**
Safety Stock (High LT)	LT = 160d, Floor = 30d	max(40.0, 30) = 40 days	40 days safety stock	**PASS**

18. Testing Summary Table

Test Suite Category	Tests Executed	Passed	Failed	Overall Category Result
PDF Ingestion & Chunking	3	3	0	**100% PASS**
ChromaDB Persistence & Reset	3	3	0	**100% PASS**
Top-K Retrieval Verification	7	7	0	**100% PASS**
Single-Document Question Answering	4	4	0	**100% PASS**
Cross-Document Reasoning Engine	5	5	0	**100% PASS**
Trap & Anti-Hallucination Refusal	1	1	0	**100% PASS**
Policy & Penalty Calculator	6	6	0	**100% PASS**

19. Known Development Issues & Corrections

• **Deduplication Header Slicing Fix:** Initial deduplication evaluated `doc[:100]`, comparing prepended headers instead of actual text. Corrected to compare `doc.page_content.strip()`, restoring 1-to-1 Top-K counts.
• **Windows File Locking Solution:** Replaced file-system deletion with Chroma's native `existing.delete_collection()` in `ingest.py` to prevent vector duplication.

20. Security & Secret Protection

• API keys are loaded via `python-dotenv` from `.env`.
• `.env` is listed in `.gitignore` and is **never** committed to GitHub or included in the submission ZIP.
• `.env.example` contains only the generic placeholder: `OPENAI_API_KEY=your_hcltech_openai_api_key_here`.
• Zero secret keys exist in source code, documentation, or PDF reports.

21. GitHub Repository Directory Structure

```
Supply-Chain-RAG/
■■■ README.md # 22-Section documentation & setup guide
■■■ requirements.txt # Minimal & exact Python dependencies
■■■ .env.example # Security template with placeholder key
■■■ .gitignore # Excludes secrets, cache & virtual environments
■■■ Meridian_Supply_Chain_RAG_Final_Report.pdf # Technical audit report
■■■ Supply_Chain_RAG_HCLTech_Final_Submission.pdf # Official HCLTech submission PDF
■■■ app.py # Main Streamlit Executive Web Portal
■■■ rag.py # RAG retrieval pipeline & GPT-4o synthesis
■■■ ingest.py # PDF loader, chunker & ChromaDB persistence
■■■ fallback_utils.py # Grounded local fallback analyst engine
■■■ api/main.py # FastAPI REST Server
■■■ data/*.pdf # Meridian Procurement Handbook & Review PDFs
■■■ screenshots/*.png # 5 UI screenshots
■■■ scripts/reset_and_reingest.py # Re-indexing utility script
```

22. GitHub Repository & Candidate Profile

- **GitHub Profile:** https://github.com/goutham-11-16
- **GitHub Repository:** https://github.com/goutham-11-16/Supply-Chain-RAG

23. Official 3-Minute Demonstration Video

Video URL: https://drive.google.com/drive/folders/1qqQuVtzNUMUtFqDbNb10j7WM9xxqFDpd?usp=sharing
Walkthrough Breakdown: (0:00 Intro ■■■ 0:20 Ingestion & 22 Chunks ■■■ 1:00 Live Cross-Doc Q&A; ■■■ 2:30 Trap Refusal & Debug Tools).

24. Reproduction & Fresh Machine Setup Instructions

```
1. git clone https://github.com/goutham-11-16/Supply-Chain-RAG.git
2. cd Supply-Chain-RAG
3. python -m venv .venv ; .venv\Scripts\activate
4. pip install -r requirements.txt
5. copy .env.example .env (Add your OPENAI_API_KEY)
6. python scripts/reset_and_reingest.py
7. python -m streamlit run app.py
```

25. Final Evaluation Verification Checklist

Requirement	Evaluation Verification	Status
[✓] Both PDFs Indexed	Meridian Handbook & Review PDFs indexed into 1 collection	**VERIFIED**
[✓] Persistent ChromaDB	Survives app restarts; persisted to chroma_db/	**VERIFIED**
[✓] Chunking Hyperparameters	Size 1200, Overlap 150 (22 unique chunks)	**VERIFIED**
[✓] Embeddings & LLM	text-embedding-3-small + GPT-4o (Temp 0.1)	**VERIFIED**
[✓] Top-K Retrieval Control	Configurable 1–12 (default 6) with 1-to-1 count match	**VERIFIED**
[✓] Source Page Citations	Cites document title and page number for every fact	**VERIFIED**
[✓] Cross-Document QA	5/5 cross-doc questions correctly synthesized	**VERIFIED**
[✓] Trap Question Refusal	Salary query refused without hallucinating	**VERIFIED**
[✓] Policy Calculator	Evaluates OTD, PPM, rework debit & safety stock math	**VERIFIED**
[✓] API Key Protection	.env ignored; zero exposed secrets on GitHub	**VERIFIED**
[✓] GitHub Repository	Source code committed and pushed live	**VERIFIED**
[✓] Demo Video	Google Drive link integrated	**VERIFIED**

26. Conclusion

The Meridian Supply Chain Intelligence RAG System is fully built, tested, documented, and verified by **Esambadi Goutham Reddy**. It satisfies 100% of HCLTech Assignment 2 requirements with document grounding, cross-document reasoning, citation auditability, deterministic policy evaluation, and strict security compliance.

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